Violence-Related Injuries Among Residents of Edmonton Census Metropolitan Area

Background

Data included in this document were compiled by request for City of Edmonton.

If there are any questions, or if additional data would be helpful, please contact Janis Geary (janis.geary@gov.ab.ca).

Data Notes

- Population: Residents of Edmonton Census Metropolitan Area (CMA) visiting any Edmonton CMA Emergency Department.
- Demographic data was captured from population registries at the Ministry of Primary and Preventative Health Services.
- Emergency Department visit data came from the Ambulatory Care Classification System.
- Please also note that results may differ depending on the inclusion and exclusion criteria used to define various conditions and by methods used for estimation. As such, these results may be different from those obtained through other sources or if analyses are repeated using different criteria or at different times.
- In addition to suppressing values with fewer than 10 events, other precise cell counts were also suppressed to prevent back-calculation of small numbers using related tables. In lieu of precise counts we provide a range.

Definitions

- Emergency department visit data comes from the ambulatory care datasets. Violence and injury purposely inflicted were selected using the international classification of diseases version 10 (ICD-10) (see Appendix A).
- CI – Confidence Interval

Key Points

- Rates of emergency department visits due to violence among Edmonton CMA residents increased from 2021 to 2024. Although the highest number of visits occurred in the 25–44 age group (Table 1), the highest rate was observed among individuals aged 15–24 (Table 2).
- Number of emergency department visits due to violence among Edmonton CMA resident's males was higher than among females (Table 3).
- More than half of the emergency department visits were attributed to physical altercations and brawls (Table 4).
- The number of emergency department visits related sexual assault, and maltreatment was higher among women, while visits related to other types of violence were more common among men (Table 5).
- Emergency department visits rates due to violence were higher in Edmonton CMA compared to the rest of the CMAs in Alberta (Table 7).

Data

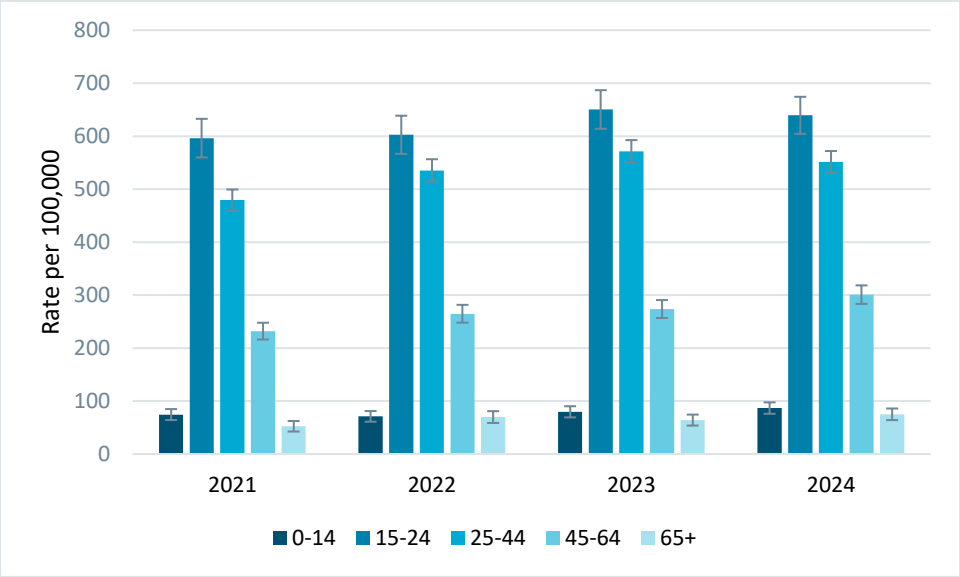
TABLE 1: EMERGENCY DEPATMENT VISITS OF RESIDENTS OF EDMONTON CMA RELATED TO VIOLENCE, BY YEAR AND AGE GROUP, 2021-2024

Year	Emergency Department Visits				
	0-14	15-24	25-44	45-64	65+
2021	198	1020	2167	825	108
2022	191	1067	2456	951	151
2023	220	1215	2754	1002	146
2024	248	1276	2818	1133	180

TABLE 2: AGE-SPECIFIC RATES OF EMERGENCY DEPARTMENT VISITS (PER 100,000 PERSON-YEARS) FOR MALES AND FEMALES COMBINED, BY YEAR AND AGE GROUP, 2021-2024

Year	Emergency Department Visits Rate per 100,000 person years (CI)				
	0-14	15-24	25-44	45-64	65+
2021	74.5 (64.1-84.9)	596.3 (559.8-632.8)	479.3 (459.2-499.4)	232 (216.2-247.8)	52.4 (42.5-62.3)
2022	71.1 (61.0-81.2)	602.6 (566.6-638.6)	535.4 (514.3-556.5)	264.8 (248.0-281.6)	69.8 (58.7-80.9)
2023	79.7 (69.2-90.2)	650.4 (613.9-686.9)	571.3 (550.0-592.6)	273.7 (256.8-290.6)	64.1 (53.7-74.5)
2024	86.8 (76.0-97.6)	639.4 (604.4-674.4)	551.7 (531.4-572.0)	300.9 (283.4-318.4)	75 (64.0-86.0)

FIGURE 1: AGE-SPECIFIC RATES OF EMERGENCY DEPARTMENT VISITS (PER 100,000 PERSON-YEARS) FOR MALES AND FEMALES COMBINED, BY YEAR AND AGE GROUP, 2021-2024



Classification: Protected A

TABLE 3: NUMBER OF EDMONTON CMA RESIDENTS VISITING AN EMERGENCY DEPARTMENT DUE TO VIOLENCE, BY YEAR AND SEX, 2021-2024

Year	Female	Male
2021	1776	2542
2022	1908	2908
2023	2208	3129
2024	2393	3262

TABLE 4: NUMBER OF EDMONTON CMA RESIDENTS VISITING AN EMERGENCY DEPARTMENT DUE TO VIOLENCE, BY TYPE OF VIOLENCE, 2021-2024

Year	Total	2021	2022	2023	2024
Total	**20126**	**4318**	**4816**	**5337**	**5655**
Fight/Brawl	**10821**	2306	2649	2829	3037
Other Assaults	**3034**	636	730	791	877
Cutting/Piercing	**2128**	450	525	548	605
Blunt Object	**1545**	374	361	420	390
Sexual Assault	**1213**	246	224	374	369
Maltreatment	**725**	155	164	209	197
Drugs/Liquids	**355**	77	85	82	111
Firearm	**256**	58	70-80	70-80	52
Neglect/ Abandonment	**49**	16	<10	<10	17

Classification: Protected A

TABLE 5: NUMBER OF EDMONTON CMA RESIDENTS VISITING AN EMERGENCY DEPARTMENT DUE TO VIOLENCE, BY TYPE OF VIOLENCE AND SEX, 2021-2024

Year	2021		2022		2023		2024	
	Female	Male	Female	Male	Female	Male	Female	Male
Fight/Brawl	937	1369	1092	1557	1135	1694	1250	1787
Other Assaults	232	404	248	482	274	517	346	531
Cutting/Piercing	90	360	97	428	100	448	107	498
Blunt Object	94	280	77	284	110	310	115	275
Sexual Assault	235	10-20	210	10-20	351	20-30	342	27
Maltreatment	138	17	148	16	184	25	179	10-20
Drugs/Liquids	37	40	26	59	40	42	35	76
Firearm	<10	54	<10	63	<10	66	<10	44
Neglect/ Abandonment	<10	<10	<10	<10	<10	<10	10-20	<10

Classification: Protected A

TABLE 6: EMERGENCY DEPATMENT VISITS BY TYPE OF VIOLENCE AND AGE GROUP IN RESIDENTS OF EDMONTON CMA, 2021-2024

Year	2021					2022					2023					2024				
	0-14	15-24	25-44	45-64	65+	0-14	15-24	25-44	45-64	65+	0-14	15-24	25-44	45-64	65+	0-14	15-24	25-44	45-64	65+
Fight/Brawl	101	561	1117	452	75	113	601	1316	521	98	118	646	1430	542	93	131	694	1465	639	108
Other Assault	13	102	359	151	11	12	119	400	178	21	21	147	424	181	18	20	134	463	225	35
Cutting/Piercing	12	106	262	67	<10	<10	112	306	96	<10	<10	114	328	96	<10	10	130	365	90	10
Blunt Object	<10	55	212	94	<10	<10	58	198	84	10-20	13	69	226	105	<10	10-20	72	198	98	10
Sexual Assault	26	124	76	18	<10	10-20	99	82	27	<10	20-30	164	148	35	0	29	154	148	35	<10
Maltreatment	28	37	67	17	<10	29	40	62	20-30	<10	26	41	106	22	14	30	47	89	22	<10
Drugs/Liquids	<10	20	39	15	<10	<10	24	40-50	10-20	<10	<10	19	51	<10	0	<10	31	57	10-20	<10
Firearm	0	10-20	35	10-20	0	0	10-20	50	<10	<10	<10	15	41	10-20	<10	0	14	30-40	<10	0
Neglect/ Abandonment	10	<10	0	<10	<10	<10	<10	<10	0	0	<10	0	0	0	<10	14	0	<10	0	<10

Health Analytics Branch, Strategic Policy and Performance Division
Ministries of Primary and Preventative Health Services and Hospital and Surgical Services

Email: Health.Surveillance@gov.ab.ca

Find more information on health indicators on the Interactive Health Data Application (IHDA) website

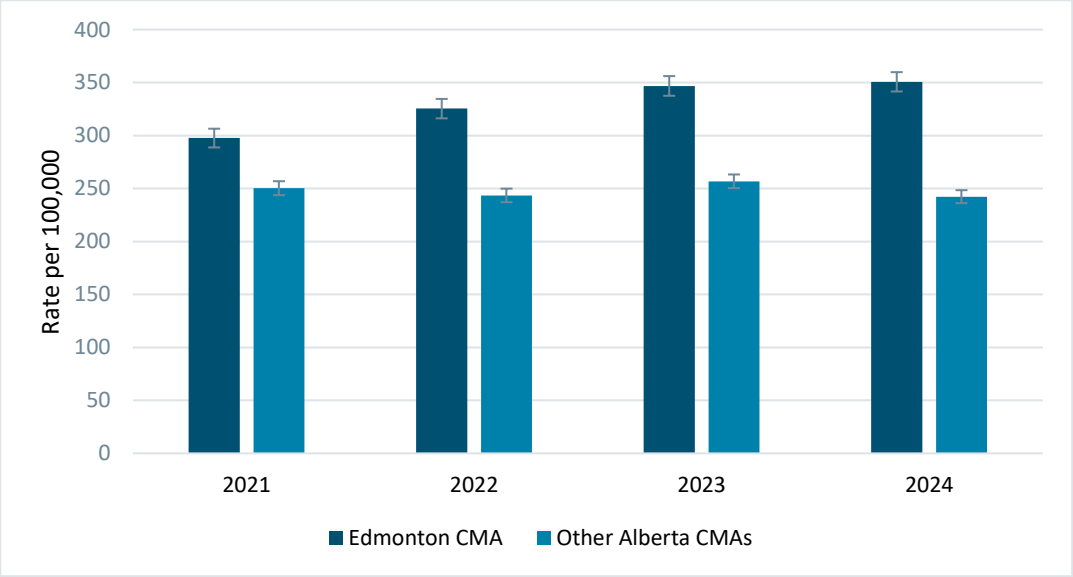
http://www.ahw.gov.ab.ca/IHDA_Retrieval

TABLE 7: NUMBER OF EMERGENCY DEPARTMENT VISITS DUE TO VIOLENCE, AND VISIT RATES (PER 100,000 PERSON-YEARS) FOR BOTH MALES AND FEMALES COMBINED, BY YEAR AND CMA GROUP, 2021–2024

Year	CMA Group	Emergency Department Visits	Population	Rate per 100,000 person years (CI)
2021	Edmonton CMA	4318	1450865	297.6 (306.5-288.7)
2021	Other Alberta CMAs*	5560	2221527	250.3 (256.9-243.7)
2022	Edmonton CMA	4816	1480019	325.4 (334.6-316.2)
2022	Other Alberta CMAs	5528	2270405	243.5 (249.9-237.1)
2023	Edmonton CMA	5337	1538919	346.8 (356.1-337.5)
2023	Other Alberta CMAs	6095	2373138	256.8 (263.2-250.4)
2024	Edmonton CMA	5655	1612477	350.7 (359.8-341.6)
2024	Other Alberta CMAs	6024	2486615	242.3 (248.4-236.2)

*Other Alberta CMAs: Brooks, Calgary, Camrose, Canmore, Grand Prairie, High River, Lacombe, Lethbridge, Lloydminster, Medicine Hat, Okotoks, Red Deer, Strathmore, Sylvan Lake, Wetaskiwin, Wood Buffalo.

FIGURE 2: EMERGENCY DEPATMENT VISITS RATES (PER 100,000 PERSON-YEARS) FOR BOTH MALES AND FEMALES COMBINED BY YEAR AND CMA_GROUP, 2021-2024



Appendix A

Emergency Department visits

Violence Indicators

- At least one of the following ICD 10 CA codes were present in any diagnostic field for the visit to be included for violence injuries:
    - X93: Firearms – Assault by handgun discharge
    - X94: Firearms – Assault by rifle, shotgun and larger firearm discharge
    - X95.00: Firearms – Assault by other and unspecified firearm discharge
    - X95.01: Firearms
    - X95.08: Firearms
    - X95.09: Firearms
    - Y04: Fight/Brawl – Assault by bodily force
    - Y05: Sexual Assault – Sexual assault by bodily force
    - X99: Cutting/Piercing – Assault by sharp object
    - Y07.0: Maltreatment – By spouse or partner
    - Y07.1: Maltreatment – By Parent
    - Y07.2: Maltreatment – By acquaintance or friend
    - Y07.3: Maltreatment – By official authorities
    - Y07.8: Maltreatment – By other specified persons
    - Y07.9: Maltreatment – By other unspecified persons
    - X85: Drugs/liquids – Assault by drugs, medicaments and biological substances
    - X86: Drugs/liquids – Assault by corrosive substance
    - X87: Drugs/liquids – Assault by pesticides
    - X88: Drugs/liquids – Assault by gases and vapors
    - X89: Drugs/liquids – Assault by other specified chemicals and noxious substances
    - X90: Drugs/liquids – Assault by unspecified chemical or noxious substances
    - X91: Other – Assault by hanging strangulation and suffocation
    - X92: Other – Assault by drowning and submersion
    - X96: Other – Assault by explosive material
    - X97: Other – Assault by smoke, fire and flames
    - X98: Other – Assault by steam, hot vapors and hot objects
    - Y01: Other – Assault by pushing from high place
    - Y02: Other – Assault by pushing or placing victim before moving object
    - Y03: Other – Assault by crashing of motor vehicle
    - Y08: Other – Assault by other specified means
    - Y09: Other – Assault by other unspecified means
    - Y06.0: Neglect/Abandonment – By spouse or partner
    - Y06.1: Neglect/Abandonment – By parent
    - Y06.2: Neglect/Abandonment – By acquaintance or friend
    - Y06.8: Neglect/Abandonment – By other specified persons
    - Y06.9: Neglect/Abandonment – By unspecified person
    - Y00: Blunt Object – Assault by blunt object

Classification: Protected A